



ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 8: HealthConnect Final Integration, Presentation & Project Showcase

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 8
Project	HealthConnect Clinic Experience Lab
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Version	1.0
Issue Date	
Submission Deadline	End of Week 8 (Sunday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 8 Submission Form

1. BUSINESS SCENARIO

HealthConnect Clinic is a fictional healthcare provider seeking to:

- Reduce missed appointments.
- Improve patient appointment attendance.
- Make better use of available appointment slots.
- Provide more effective administrative support to patients.

During **Week 4**, interns established their understanding of the HealthConnect problem, reviewed the available resources and defined their track-specific responsibilities.

During **Week 5**, interns moved from planning into practical implementation.

During **Week 6**, interns advanced their work, integrated components across tracks and validated their outputs.

During **Week 7**, interns tested their work, identified issues, refined their components and performed end-to-end validation.

The project now moves into its final stage:

Final Integration → Project Readiness → Presentation → Showcase

The objective is to demonstrate how the individual contributions from the different professional tracks come together to form a coherent HealthConnect solution.

The overall project question remains:

How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

2. PROJECT CONTINUITY

The HealthConnect project follows the established project journey:

Week	Project Stage
Week 4	Problem Understanding → Resource Review → Solution Planning
Week 5	Analysis → Development → Initial Implementation
Week 6	Integration → Advanced Development → Validation
Week 7	Testing → Refinement → End-to-End Validation
Week 8	Final Integration → Presentation

The Week 6 assignment explicitly established this progression and stated that Week 8 would be the final integration and presentation stage.

Important

You must continue from your previous HealthConnect work.

Do not:

- Start a new HealthConnect project.
- Select a new dataset.
- Discard your previous work.
- Repeat the complete Week 5, 6 or 7 assignments.
- Create an unrelated project.
- Present work that does not represent your actual contribution.
- Claim that a component was completed when it was not.

Week 8 should demonstrate the **evolution of your work from Week 4 through Week 7 into a final project contribution.**

3. WEEK 7 → WEEK 8 TRANSITION

Before beginning your final work, briefly document:

1. Your main Week 7 output.
2. The most important testing result.
3. The major issue discovered during testing.
4. What was refined or improved.
5. Which component is now considered ready.
6. What still needs to be integrated.
7. Which track(s) are involved in the final integration.
8. What you intend to demonstrate during the final presentation.

This should be concise.

Do not reproduce your Week 7 report.

The purpose is to demonstrate how testing and refinement led to the final HealthConnect solution.

4. WEEK 8 ASSIGNMENT OBJECTIVE

The objective of Week 8 is to **finalize, integrate and professionally present the HealthConnect solution developed throughout the Experience Lab.**

By the end of Week 8, interns should be able to:

- Integrate their final track-specific contribution into the broader HealthConnect solution.
- Demonstrate how different project components work together.
- Resolve or clearly document important remaining issues.
- Present evidence supporting their decisions and outputs.
- Communicate technical or analytical work to a professional audience.
- Explain the business value of their contribution.
- Demonstrate meaningful cross-track collaboration.
- Produce a professional final project portfolio.
- Present the completed HealthConnect solution professionally.

5. WEEK 8 LEARNING OBJECTIVES

By completing this assignment, interns should be able to:

- Translate technical work into a business-focused solution.
- Integrate multidisciplinary project components.
- Communicate project outcomes clearly.
- Demonstrate evidence-based decision-making.
- Explain limitations and risks professionally.
- Present technical findings to both technical and non-technical audiences.
- Collaborate effectively within an HC-POD.
- Demonstrate individual ownership within a shared project.

- Prepare professional project documentation.
- Build a portfolio-ready project case study.
- Reflect on lessons learned throughout the project.

6. PROJECT RESOURCES

Continue using the approved HealthConnect project resources:

6.1 HealthConnect Appointment Dataset

HealthConnect_Appointment_Data.csv

6.2 HealthConnect Data Dictionary

HealthConnect_Data_Dictionary.xlsx

6.3 HealthConnect Clinic Knowledge Base

HealthConnect_Clinic_Knowledge_Base.docx

These remain the approved project resources.

Do not overwrite the original resources. Processed or transformed files must remain separate and properly documented.

For Generative AI work:

Responses must remain within the approved HealthConnect Knowledge Base and must not invent clinic or medical information.

7. PART 1 — FINAL INTEGRATION READINESS

Before completing the final presentation, each intern must establish that their work is ready for integration.

Document:

Requirement	Your Response
What is my final component?	
What problem does it address?	
What is its current status?	
What did I improve during Week 7?	

Requirement

Your Response

Which other track(s) does it depend on?

Which track(s) depend on my output?

What output will I provide?

What output do I need from another track?

What evidence demonstrates readiness?

What limitations remain?

8. TRACK-SPECIFIC ASSIGNMENTS

A. PROJECT MANAGEMENT TRACK

Week 8: Final Project Integration, Readiness & Presentation Coordination

The Project Management track is responsible for helping coordinate the final stage of the HealthConnect project.

Tasks

1. Review the overall project status.
2. Review outstanding issues from Week 7.
3. Confirm the status of major project components.
4. Identify unresolved dependencies.
5. Coordinate final cross-track integration.
6. Confirm that each track has its required final output.
7. Update the project timeline/status.
8. Coordinate the final integration walkthrough.
9. Develop the final presentation structure.
10. Coordinate presentation responsibilities.
11. Document major project decisions.
12. Document unresolved risks or limitations.
13. Confirm final project readiness.
14. Coordinate final submission requirements.
15. Prepare a concise project closure/status report.
16. Coordinate the HC-POD final presentation.

Week 8 Output

HealthConnect Final Project Management & Readiness Package

Include:

- Final project status report.
- Updated project plan/timeline.
- Final workstream status.
- Outstanding issues/dependency register.
- Final risk register.
- Final decision log.
- Final integration record.
- Presentation coordination plan.
- Project readiness assessment.
- Project closure summary.

B. DATA ANALYTICS TRACK

Week 8: Final Analytics, Dashboard & Business Insights

The Data Analytics track should transform the analytical work completed during Weeks 5–7 into a **final decision-support output**.

Tasks

1. Review the validated findings from Week 7.
2. Confirm the final KPIs.
3. Validate important dashboard figures.
4. Ensure visualizations are accurate and understandable.
5. Finalize the dashboard or analytical report.
6. Present the most important attendance/no-show patterns.
7. Identify the most important business insights.
8. Translate findings into actionable recommendations.
9. Explain how the findings support HealthConnect decisions.
10. Provide relevant final findings to other tracks.
11. Document important analytical limitations.
12. Prepare the analytics section of the final presentation.
13. Ensure the dashboard/report is professionally presented.
14. Prepare a concise executive summary of the analytical findings.

Week 8 Output

HealthConnect Final Analytics & Decision Support Package

Include:

- Final dashboard/report.
- Final KPIs.
- Validated findings.

- Key visualizations.
- Business insights.
- Recommendations.
- Analytical limitations.
- Cross-track contribution.
- Executive summary.
- Presentation slides/materials.

C. DATA SCIENCE TRACK

Week 8: Final Model Selection, Documentation & Presentation

The Data Science track should move from the tested/refined candidate model to a **final documented modelling outcome**.

Tasks

1. Review the Week 7 model testing results.
2. Confirm the final candidate model.
3. Compare final performance against the baseline.
4. Present the most relevant evaluation metrics.
5. Explain important model strengths and weaknesses.
6. Document important modelling decisions.
7. Explain the business suitability of the model.
8. Incorporate relevant validated findings from Data Analytics.
9. Communicate final model requirements to ML Engineering.
10. Document model limitations and risks.
11. Clearly state what the model can and cannot be used for.
12. Prepare the model section of the final presentation.
13. Prepare a concise model summary for non-technical stakeholders.
14. Ensure the final notebook/report is professionally organized.

Week 8 Output

HealthConnect Final Model & Data Science Package

Include:

- Final candidate model.
- Model evaluation.
- Baseline vs final comparison.
- Error analysis summary.
- Feature/model decisions.
- Business interpretation.
- Limitations and risks.
- Cross-track contribution.
- Final model documentation.
- Presentation materials.

The original HealthConnect project plan also identifies Week 8 as the stage for the final report/presentation, cross-track walkthrough and final presentation.

D. MACHINE LEARNING ENGINEERING TRACK

Week 8: Final ML Pipeline Integration, Documentation & Readiness

The ML Engineering track should prepare the integrated ML workflow for final demonstration.

Tasks

1. Review Week 7 pipeline testing results.
2. Resolve critical pipeline issues where possible.
3. Confirm compatibility between preprocessing and the final model.
4. Confirm expected inputs and outputs.
5. Validate the end-to-end ML workflow.
6. Confirm configuration and dependencies.
7. Ensure reproducibility.
8. Review error handling and logging.
9. Organize the repository professionally.
10. Update README.md.
11. Document how the pipeline works.
12. Document known limitations.
13. Coordinate with Data Science regarding the final model.
14. Prepare the ML workflow for demonstration.
15. Prepare the ML Engineering section of the final presentation.

Important

Week 8 does **not automatically mean production deployment**.

The focus is on **final integration, demonstration readiness, documentation and presentation**, consistent with the HealthConnect project journey.

Week 8 Output

HealthConnect Final ML Pipeline & Engineering Package

Include:

- Final integrated pipeline.
- Project/repository structure.
- Requirements/dependencies.
- Configuration documentation.
- Testing summary.
- Error-handling evidence.
- Reproducibility information.
- README.md.
- Known limitations.
- Final integration evidence.

- Presentation materials.

E. GENERATIVE AI TRACK

Week 8: Final AI Assistant Integration, Safety & Demonstration

The GenAI track should transform the refined assistant from Week 7 into a **final demonstrable HealthConnect administrative support component**.

Tasks

1. Review Week 7 assistant testing results.
2. Confirm final supported use cases.
3. Confirm out-of-scope use cases.
4. Finalize the prompt/system instructions.
5. Confirm Knowledge Base integration.
6. Test representative final scenarios.
7. Demonstrate handling of unsupported questions.
8. Demonstrate escalation behavior.
9. Confirm that the assistant does not invent clinic information.
10. Confirm consistency of responses.
11. Document known limitations.
12. Prepare a final demonstration workflow.
13. Coordinate with Project Management regarding the final demo.
14. Prepare the GenAI section of the final presentation.

Final demonstration should show examples such as:

- A normal supported request.
- An ambiguous request.
- An unavailable-information request.
- An unsupported request.
- An escalation scenario.

Week 8 Output

HealthConnect Final AI Assistant & Safety Package

Include:

- Final prompt/instruction set.
- Knowledge Base approach.
- Final test results.
- Safety validation.
- Escalation behavior.
- Sample outputs.
- Demonstration workflow.
- Limitations.
- Cross-track contribution.
- Final presentation materials.

9. MANDATORY HC-POD FINAL INTEGRATION

This is one of the **most important parts of Week 8**.

The HC-POD should now move from individual track contributions into a **coherent multidisciplinary HealthConnect solution**.

Week 6 established that collaboration must involve an actual exchange and use of outputs—not communication alone.

By Week 8, the HC-POD should demonstrate the **final connection between the components**.

Each intern must document:

1. Track collaborated with.
2. Dependency.
3. Output received.
4. Output provided.
5. Final integration activity.
6. What changed.
7. How the change improved the solution.
8. Evidence.
9. Contribution to the final presentation.

Example

Data Analytics → Data Science

Analytics provides validated no-show patterns → Data Science explains how those findings influenced model interpretation.

Data Science → ML Engineering

Final candidate model + requirements → ML Engineering integrates the model into the final workflow.

ML Engineering → Data Science

Pipeline requirements/constraints → Data Science confirms model compatibility.

GenAI → Project Management

Final assistant capabilities, limitations and demonstration requirements → PM incorporates these into final project presentation.

Project Management → All Tracks

Coordinates final integration, outstanding dependencies, presentation sequence and project readiness.

10. HEALTHCONNECT FINAL INTEGRATION MATRIX

Track	Final Contribution
Data Analytics → Data Science	Validated analytical findings support final model interpretation
Data Science → ML Engineering	Final model and technical requirements support final ML workflow
ML Engineering → Data Science	Pipeline requirements support final model integration
Data Analytics → PM	Final insights support business recommendations and project decisions
GenAI → PM	Final assistant capabilities, limitations and demo requirements
PM → All Tracks	Coordinates final integration, readiness, presentation and dependencies
All Tracks → All Tracks	Relevant information required for the complete HealthConnect solution

These relationships build on the integration approach already established in Week 6.

11. FINAL END-TO-END INTEGRATION WALKTHROUGH

Each HC-POD should conduct a final walkthrough demonstrating:

1. The Problem

What problem is HealthConnect trying to solve?

2. The Data

What data/resources were used?

3. The Analytics

What did the data reveal?

4. The Predictive Component

How does Data Science contribute?

5. The ML Engineering Component

How is the model/process organized into a usable technical workflow?

6. The GenAI Component

How does the AI assistant support the healthcare administrative experience?

7. The Project Management Layer

How were the different workstreams coordinated?

8. The Final Solution

How do these components work together?

9. Business Value

How could the proposed solution help HealthConnect?

10. Limitations

What can the solution **not** currently do?

12. FINAL PROJECT PRESENTATION — INDIVIDUAL VIDEO SUBMISSION

Each intern must prepare and submit an **individual recorded video presentation of 5–10 minutes**.

The video should demonstrate the intern's understanding of the HealthConnect project, their individual track-specific contribution, their cross-track collaboration and how their work contributed to the final integrated solution.

Important

Although HealthConnect is a multidisciplinary project and interns work within HC-PODs, **the final presentation video must be submitted individually by every intern**.

Interns must not simply submit a video showing other team members presenting.

Each intern is expected to personally explain their contribution.

The 5–10-minute video should cover:

1. Introduction — 30–60 seconds

- Your name.
- Internship track.

- HealthConnect project.
- The problem being addressed.

2. Your Track Contribution — 2–3 minutes

- Your assigned responsibility.
- What you developed or produced.
- Key methods/tools used.
- Important results or outcomes.

3. Cross-Track Collaboration — 1–2 minutes

- Which track(s) you worked with.
- What you exchanged.
- How you used the other track's output.
- What changed because of the collaboration?

4. Testing, Refinement & Final Outcome — 1–2 minutes

- What you tested.
- What you improved.
- Important findings/results.
- Current limitations.

5. Overall HealthConnect Solution — 1–2 minutes

- Explain how your work connects with the other components.
- Explain the overall business value of the solution.
- State how your contribution supports the HealthConnect objective.

6. Closing — 30–60 seconds

- Key takeaway.
- What you learned.
- What could be improved in a future version.

13. FINAL HC-POD INTEGRATION WALKTHROUGH

In addition to the **individual video submission**, each HC-POD should complete a final integration walkthrough.

The purpose of the walkthrough is to demonstrate how:

Data Analytics + Data Science + ML Engineering + Generative AI + Project Management

come together as one HealthConnect solution.

The HC-POD should demonstrate:

- The original business problem.

- The different track contributions.
- How the components connect.
- How information/outputs moved between tracks.
- How testing and refinement improved the solution.
- The final proposed HealthConnect workflow.
- Key results.
- Limitations and recommendations.

Important distinction

HC-POD = collaborative project integration

Individual video = individual assessment and presentation

Therefore, every intern must be able to explain their own work even if another member of the HC-POD helped with the integration.

14. ASSESSMENT CRITERIA

Assessment Criteria	Weight
Application of Previous Project Work	10%
Final Track-Specific Quality	20%
Final Integration & Solution Coherence	20%
Testing, Validation & Evidence	10%
Problem-Solving & Decision-Making	10%
HC-POD Cross-Track Collaboration	10%
Documentation & Portfolio Quality	10%
Final Presentation & Communication	10%
Total	100%

Professional Development Requirement

As part of your professional development, you should:

- Update your existing GitHub repository.
- Update your `README.md`.
- Document your analytical process.
- Present your dashboard professionally.
- Share your learning and project progress on LinkedIn and X where appropriate.
- Tag AnalystLab Africa where applicable.
- Use **#AnalystLabAfrica** where appropriate.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.